

Essential Summary

Generalized Function Projective Synchronization of Two Different Hyperchaotic Systems with Unknown Parameters

Zhenbo Li, Xiaoshan Zhao

Nonlinear Analysis: Real World Applications, Vol. 12, No. 5, 2607–2615 (2011)

DOI: 10.1016/j.nonrwa.2011.03.009

This is an author-prepared essential summary, not the original paper.
Please cite the published version.

1 Background and Motivation

Projective synchronization is attractive for secure communications because its proportional feature enables fast communication. The concept evolved through several generalizations: **projective synchronization** (Mainieri & Reháček, 1999, partially linear systems) → **modified projective synchronization** (MPS, constant scaling *matrix*) → **function projective synchronization** (FPS, scaling *function*) → **modified function projective synchronization** (MFPS, scaling function *matrix*) → **generalized function projective synchronization** (GFPS).

This paper studies **GFPS between two different hyperchaotic systems whose parameters are completely unknown**. Combining adaptive control theory with an **antisymmetric structure**, the authors design an extended adaptive controller that is (i) more generalized than the controller in Yu & Li (2010) and (ii) simpler than that in Du et al. (2009). Unknown parameters are simultaneously estimated.

2 GFPS Scheme

2.1 Drive–Response Systems

Drive system with unknown parameters Θ :

$$\dot{x} = f(x) + \Phi(x)\Theta, \quad (1)$$

Response system with unknown parameters Ω and controller U :

$$\dot{y} = g(y) + \Psi(y)\Omega + U. \quad (2)$$

2.2 Error Definition

The GFPS error is defined relative to a **scaling function matrix** $h(x) = \text{diag}\{h_1(x), h_2(x), \dots, h_n(x)\}$:

$$\boxed{e = y - h(x)x, \quad \dot{e} = g(y) + \Psi(y)\Omega - J(f(x) + \Phi(x)\Theta) + U} \quad (3)$$

where $J = d(h(x)x)/dx$ is the Jacobian of the scaled state.

Remark (hierarchy): If all $h_i(x)$ are equal, GFPS → FPS. If all h_i are constants, GFPS → MPS. If $h \equiv 0$, the problem reduces to chaos control. Thus GFPS unifies many existing schemes.

3 Controller Design via Antisymmetric Structure

3.1 Stability Lemma

Theorem 1. For $\dot{x} = B(x)x$ with $B(x) = B_1(x) + B_2$, if

- (i) $B_1^T(x) = -B_1(x)$ (antisymmetric),
- (ii) $B_2 = \text{diag}\{b_1, \dots, b_n\}$ with $b_i < 0$,

and the invariant set contains only the origin, then the system is asymptotically stable.

Proof. With $V = \frac{1}{2}x^T x$, the antisymmetry of B_1 cancels the symmetric part:

$$\dot{V} = \frac{1}{2}(x^T B^T x + x^T B x) = x^T B_2 x < 0. \quad (4)$$

3.2 Main Controller (Theorem 2)

GFPS is achieved by the controller and parameter update laws:

$$U = J(f(x) + \Phi(x)\hat{\Theta}) - g(y) - \Psi(y)\hat{\Omega} + Ke \quad (5)$$

$$\dot{\hat{\Theta}} = -J^T \Phi^T(x)e + \tilde{\Theta}, \quad \dot{\hat{\Omega}} = \Psi^T(y)e + \tilde{\Omega}, \quad (6)$$

where $\hat{\Theta}, \hat{\Omega}$ are parameter estimates, $\tilde{\Theta} = \Theta - \hat{\Theta}$, $\tilde{\Omega} = \Omega - \hat{\Omega}$, and K satisfies the assumptions of Theorem 1 (antisymmetric off-diagonal + negative diagonal).

Proof. Choose $V = \frac{1}{2}(e^T e + \tilde{\Theta}^T \tilde{\Theta} + \tilde{\Omega}^T \tilde{\Omega})$. Substituting the controller and update laws yields

$$\dot{V} = \frac{1}{2}(e^T K^T e + e^T K e) - \tilde{\Theta}^T \tilde{\Theta} - \tilde{\Omega}^T \tilde{\Omega} < 0, \quad (7)$$

which guarantees global asymptotic stability of the error system and convergence of parameter estimates.

Remarks. The controller is simpler than Du et al. (2009) and more generalized than Yu & Li (2010), whose controller is a special case. The scaling functions may be periodic or polynomial functions.

4 Application: Hyperchaotic Chen $\rightarrow$ Hyperchaotic Lorenz

The drive is a hyperchaotic Chen system and the response a hyperchaotic Lorenz system, both with fully unknown parameters:

$$\dot{x} = \begin{pmatrix} -ax_1 + ax_2 + x_4 \\ dx_1 + cx_2 - x_1x_3 \\ -bx_3 + x_1x_2 \\ rx_4 + x_2x_3 \end{pmatrix}, \quad \dot{y} = \begin{pmatrix} -a_1y_1 + a_1y_2 + y_4 \\ b_1y_1 - y_2 - y_1y_3 \\ -c_1y_3 + y_1y_2 \\ d_1y_4 - y_1y_3 \end{pmatrix} + U. \quad (8)$$

True parameter values: $a = 35$, $b = 3$, $c = 12$, $d = 7$, $r = 0.08$; $a_1 = 10$, $b_1 = 28$, $c_1 = \frac{8}{3}$, $d_1 = 1.3$.

4.1 Case 1: Periodic Scaling Functions

$$h_i(x_i) = d_{i1} \sin(d_{i2}x_i + d_{i3}) + d_{i4}, \quad i = 1, 2, 3, 4. \quad (9)$$

Figures 1–2 show that the eight unknown parameters adapt to their true values and the synchronization errors converge to zero.

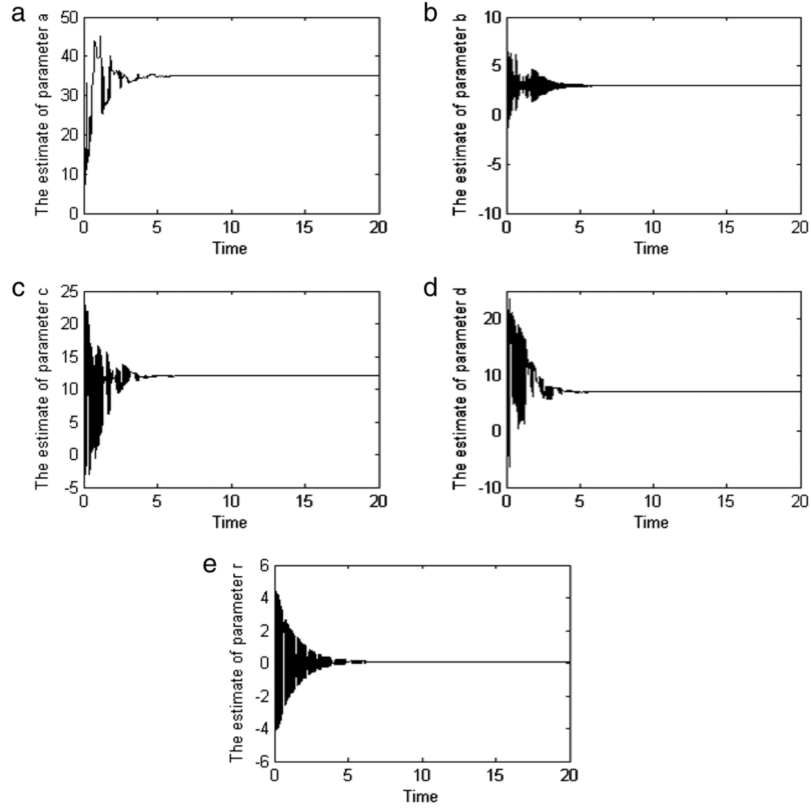


Figure 1: Estimation of the unknown parameters of the drive system (hyperchaotic Chen) under the parameter update law and periodic scaling functions.

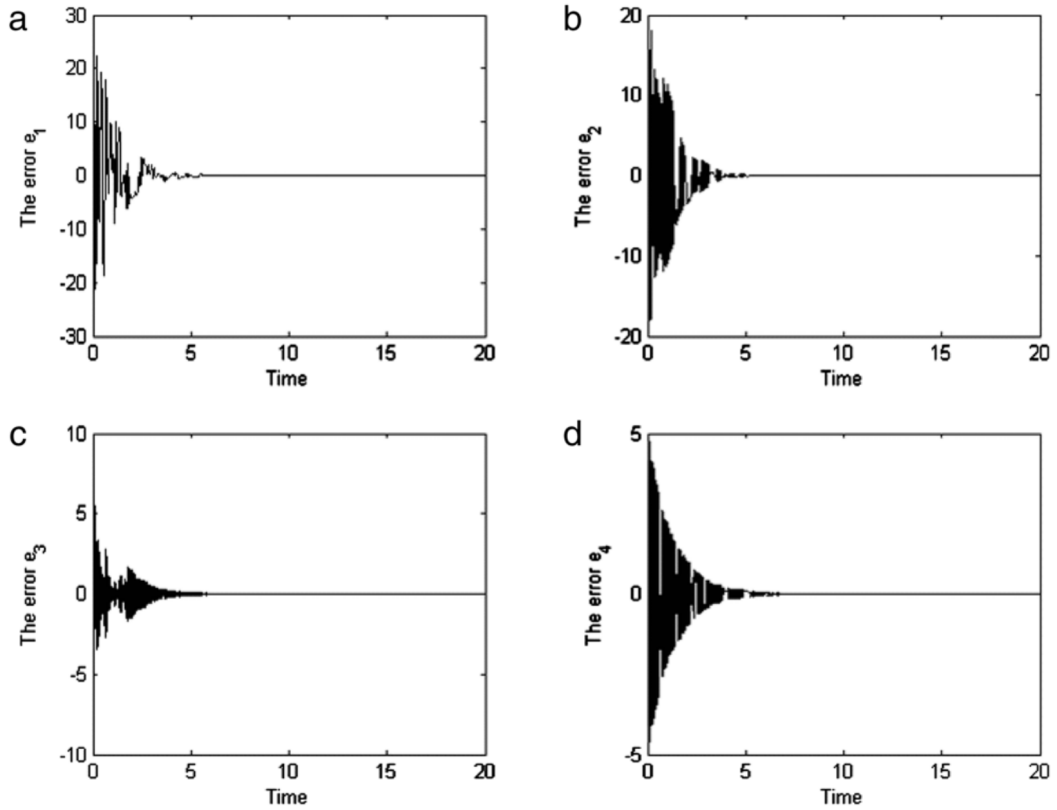


Figure 2: Time response of the generalized function projective synchronization errors between the two hyperchaotic systems with periodic scaling functions. The errors converge to zero as $t \rightarrow \infty$.

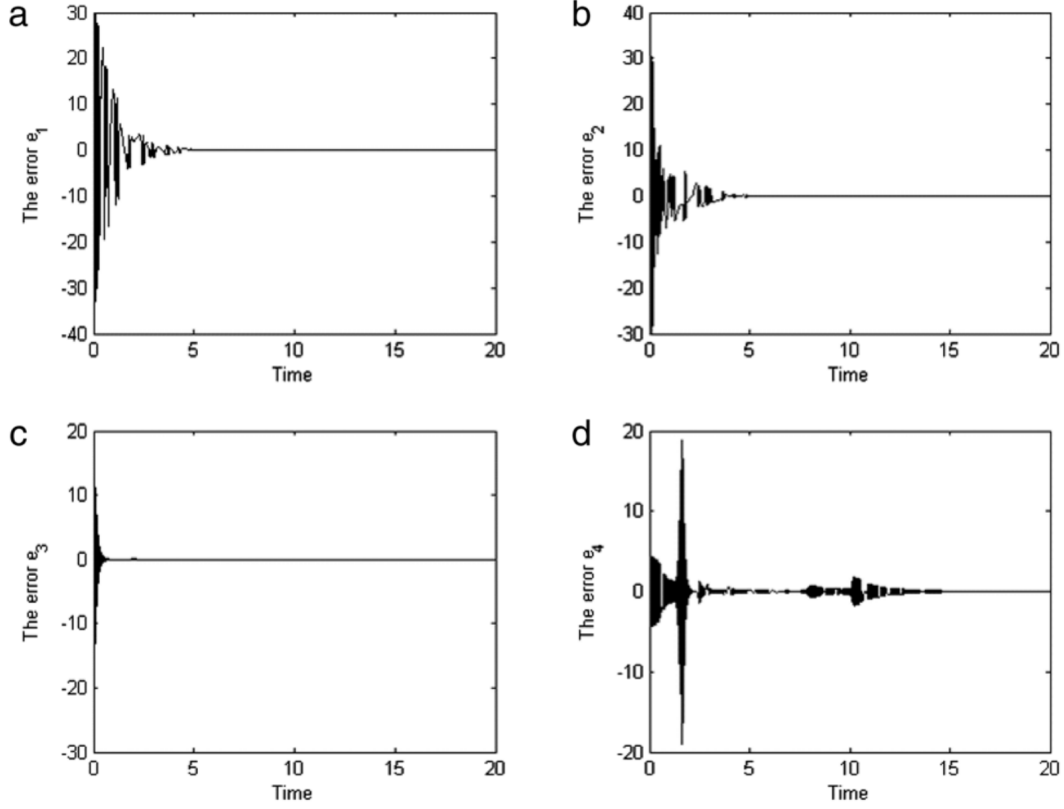


Figure 3: Time response of the generalized function projective synchronization errors with polynomial scaling functions.

4.2 Case 2: Polynomial Scaling Functions

$$h_i(x) = d_{i1}x_i^2 + d_{i2}x_i + d_{i3}, \quad i = 1, 2, 3, 4. \quad (10)$$

The corresponding parameter-estimation and error-response results confirm that GFPS is achieved and the unknown parameters are identified equally well.

5 Key Findings

1. **Unified GFPS framework:** The scaling function matrix $h(x)$ generalizes FPS (scalar function), MPS (constant matrix), and chaos control ($h \equiv 0$) as special cases.
2. **Simpler, more general controller:** Combining adaptive control with the antisymmetric structure eliminates the need for complex matrix design. The controller is simpler than Du et al. (2009) and more general than Yu & Li (2010).
3. **Parameter estimation for hyperchaotic systems:** All eight unknown parameters of the Chen and Lorenz hyperchaotic systems are simultaneously estimated while synchronization is achieved — handled naturally by the adaptive update laws.
4. **Complicated scaling functions:** Periodic (sine) and polynomial scaling functions, which had not been discussed in prior literature, are both demonstrated to work.
5. **Lyapunov-based guarantee:** The antisymmetric structure ensures $\dot{V} = e^T B_2 e < 0$ (see Eqs. (4) and (7)), giving a rigorous global asymptotic stability proof.

6 Significance

This paper is the author's second major work in chaos synchronization (after the 2011 *Acta Physica Sinica* paper on modified active control for generalized projective synchronization). Together, these two papers establish the "chaos synchronization" thread in the author's research trajectory. The key methodological idea — constructing controllers via an antisymmetric structure validated by Lyapunov theory rather than by algebraic eigenvalue criteria — mirrors the same Lyapunov-centric design philosophy later applied to analytical perturbation methods (GHFP/MGHFP series).

Author: Zhenbo Li (lizhenbo@usc.edu.cn)

ORCID: 0000-0003-2290-971X

University of South China, School of Mathematics and Physics
Hunan Key Laboratory of Mathematical Modeling and Scientific Computing